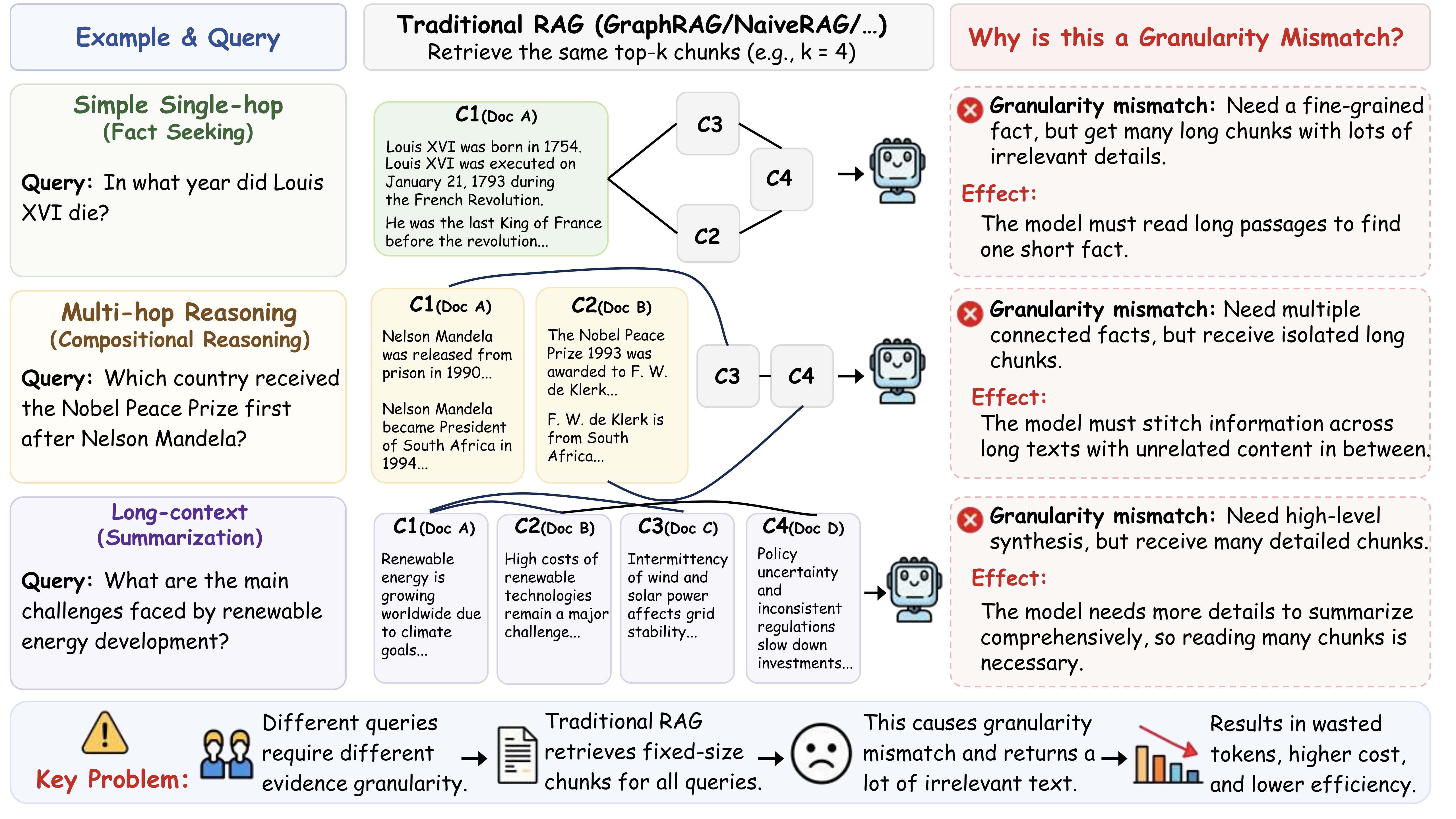

Example & Query
Traditional RAG (GraphRAG/NaiveRAG/...)
Retrieve the same top-k chunks (e.g., k = 4)
Why is this a Granularity Mismatch?
Simple Single-hop
(Fact Seeking)
Query: In what year did Louis XVI die?
C1(Doc A)
Louis XVI was born in 1754. Louis XVI was executed on January 21, 1793 during the French Revolution.
He was the last King of France before the revolution...
C3
C4
C2
Granularity mismatch: Need a fine-grained fact, but get many long chunks with lots of irrelevant details.
Effect:
The model must read long passages to find one short fact.
Multi-hop Reasoning
(Compositional Reasoning)
Query: Which country received the Nobel Peace Prize first after Nelson Mandela?
C1(Doc A)
Nelson Mandela was released from prison in 1990...
Nelson Mandela became President of South Africa in 1994...
C2(Doc B)
The Nobel Peace Prize 1993 was awarded to F. W. de Klerk...
F. W. de Klerk is from South Africa...
C3
C4
Granularity mismatch: Need multiple connected facts, but receive isolated long chunks.
Effect:
The model must stitch information across long texts with unrelated content in between.
Long-context
(Summarization)
Query: What are the main challenges faced by renewable energy development?
C1(Doc A)
Renewable energy is growing worldwide due to climate goals...
C2(Doc B)
High costs of renewable technologies remain a major challenge...
C3(Doc C)
Intermittency of wind and solar power affects grid stability...
C4(Doc D)
Policy uncertainty and inconsistent regulations slow down investments...
Granularity mismatch: Need high-level synthesis, but receive many detailed chunks.
Effect:
The model needs more details to summarize comprehensively, so reading many chunks is necessary.
Key Problem:
Different queries require different evidence granularity.
Traditional RAG retrieves fixed-size chunks for all queries.
This causes granularity mismatch and returns a lot of irrelevant text.
Results in wasted tokens, higher cost, and lower efficiency.